



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

VISION SENSORS [CSE-4016]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse the role of Modulation Transfer Function (MTF) in evaluating the performance of an optical system. (5)
A)
B) Summarize how aerial cameras contribute to remote sensing applications. (5)
- 2) Categorize different types of lens aberrations and explain their impact on image quality. (5)
A)
B) List and explain the principles involved in image reconstruction in Computed Tomography and analyse their contribution to the final image clarity. (5)
- 3) Apply the concept of spectral resolution to differentiate between hyperspectral and multispectral images (5)
A)
B) Distinguish between image acquisition techniques using a single sensor and sensor arrays. (5)
- 4) Infer how the sampling in the wavelength domain influences the representation of colour in digital images (5)
A)
B) Compare the performance of CCD and CMOS sensors (5)
- 5) Illustrate the working of CCD sensors (3)
A)
B) Explain the concept of emissivity in Infrared Thermography (2)
C) List and explain the principles involved in image reconstruction in Computed Tomography and analyse their contribution to the final image clarity. (5)

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